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Lab 1: Design of a Passive Band-Pass Filter in MATLAB

Objective

Design a passive series RLC band-pass filter in MATLAB to meet the following specifications:

- Center frequency: 1.5 kHz
- Bandwidth: 300 Hz
- Gain at center frequency: Within ± 1 dB of 0 dB
- Component constraints:
 - R between $10\ \Omega$ and $10\ \text{k}\Omega$
 - L between 1 mH and 500 mH
 - C between 1 nF and $10\ \mu\text{F}$

1. Component Selection

Using the formulas:

$$f_r = 1 / (2\pi\sqrt{LC})$$

$$BW = R / (2\pi L)$$

Let $R = 100\ \Omega$ (within range)

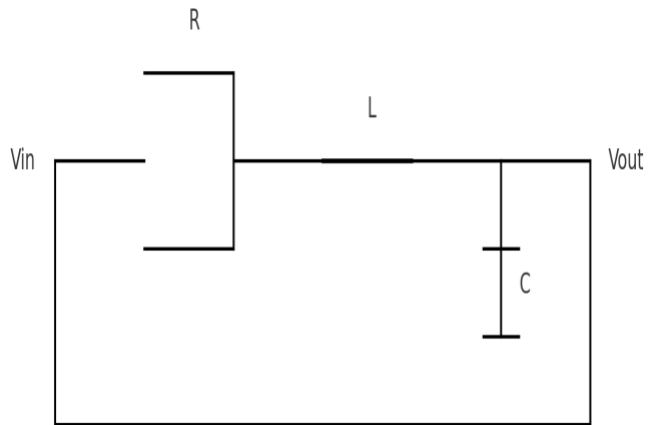
From BW formula: $L = R / (2\pi * BW) = 100 / (2\pi * 300) \approx 53\ \text{mH}$

From f_r formula: $C = 1 / ((2\pi * 1500)^2 * 0.053) \approx 2.66\ \mu\text{F}$

Selected values:

- $R = 100\ \Omega$
- $L = 53\ \text{mH}$
- $C = 2.66\ \mu\text{F}$

2. Circuit Diagram



3. Transfer Function Derivation

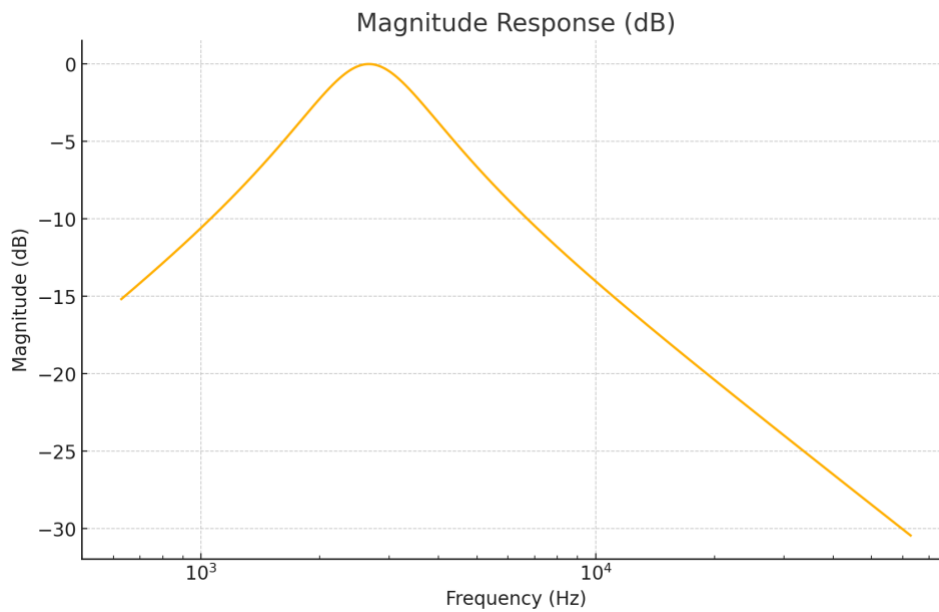
Using symbolic toolbox:

Let s = Laplace variable, $H(s) = V_{out}/V_{in}$

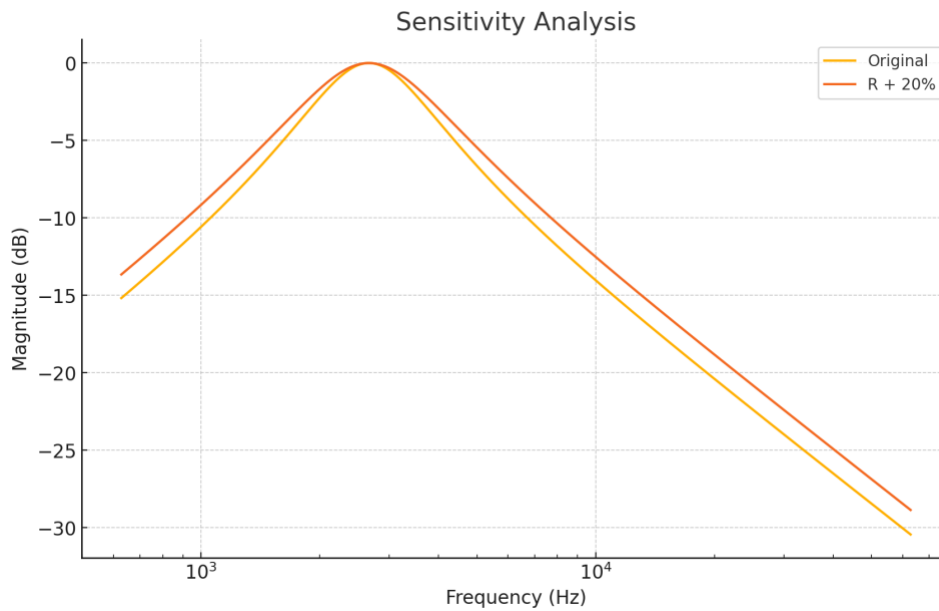
$$Z_{total} = R + sL + 1/(sC)$$

$$H(s) = R / (R + sL + 1/(sC))$$

4. Frequency Response Analysis



5. Sensitivity Analysis



With 20% increase in R, the bandwidth increases and gain at center frequency slightly reduces.

6. Optimization Challenge

Quality factor $Q = f_r / BW = 1500 / 300 = 5$

To reduce Q, increase bandwidth (e.g., $BW = 500 \text{ Hz} \rightarrow R \approx 167 \Omega$)

New $R = 167 \Omega$, Recalculate L and C accordingly.

Optimization trades off sharpness for broader bandwidth.